
UNIT 6 MATRIX ALGEBRA

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6.0 INTRODUCTION

The concept of matrix has been used for thousands of years in Mesopotamia, India and China for various purposes including solving systems of simultaneous equations. Analysis of many real-life problems and relationships among entities often requires inclusion of a large number of interrelated variables into the model. In such cases matrix is used for solving the problems. The knowledge of matrices has become necessary for the individuals working in different branches of science, technology, commerce, management and social sciences. In this unit, we introduce the concept of matrices and its basic properties. Also, we will learn areas of applications of matrices.

6.1 OBJECTIVES

After going through this unit, you should be able to

- explain the concept of matrix,
- define a matrix and give examples of matrices,
- explain the types of matrices,

- explain and use related concepts of submatrices and minors, orthogonal matrix, symmetric and skew-symmetric matrices, idempotent Matrix, adjoint and reciprocal matrices, trace of a matrix, transpose of matrix, non-singular matrix, inverse of a matrix;
- find inverse of a non-singular matrix,
- calculate rank of a matrix,
- use partition matrix in carrying out efficiently matrix operations, and
- explain applications of matrix.

6.2 DEFINITION OF A MATRIX

In real life, we do not have to deal with a single magnitude but with a set of magnitudes simultaneously. In analysis we treat such a set of magnitudes as a single entity or a single object of thought, abstracted by numbers. Let us consider a problem where we treat a set as single entity and a number of such sets as a single system. Later in this unit we will examine how the fundamental operations of arithmetic can be suitably defined to deal with such a system.

Suppose there are 3 individuals A, B, C in a family and that

A has a set of 3 pants, 3 shirts and 1 tie.

B has a set of 5 pants, 5 shirts and 2 ties.

C has a set of 6 pants, 8 shirts and no tie.

We can arrange the data in following convenient system:

Individuals ↓		Pants (P)	Shirts (S)	Ties (T)		
A	→	3	3	1	→	1 st Row
B	→	5	5	2	→	2 nd Row
C	→	6	8	0	→	3 rd Row
		↓	↓	↓		
		1 st	2 nd	3 rd		
		Col.	Col.	Col.		

This system comprises 3 rows and 3 columns (note that, by convention, the number of rows are written first and then the number of columns afterwards). We have written belongings of A in the first row, belongings of B in the second row and the belongings of C in the third row. Clearly, 1st column gives us the numbers of pants that A, B and C respectively have, while second column gives us shirts of A, B, and C respectively; and similarly, third column gives the number of ties of A, B, C respectively.

Let us take another example to arrive at the definition of a matrix:

Suppose there are three girls, “Kavita, Shrikant and Tanu.” Kavita has 9 hundred rupees notes, 4 fifty rupees notes, and 5 ten rupees notes. Shrikant has

17 hundred rupees notes, 6 fifty rupees notes and one ten rupee note. Tanu has 8 hundred rupees notes, 3 fifty rupees notes and 2 ten rupees notes.

This information can be represented as:

		Column1	Column2	Column3	
		↓	↓	↓	
		Rs. 100	Rs. 50	Rs. 10	
		Notes	Notes	Notes	
Row1	→	Kavita	9	4	5
Row2	→	Shrikant	17	6	1
Row3	→	Tanu	8	3	2

This is an arrangement of 9 (3×3) numbers in 3 rows and 3 columns. Such an arrangement is **nothing but a matrix**.

The arrangement of numbers written in such a particular form of rows and columns and enclosed by [], the square brackets, or (), the large parentheses, is called a '**matrix**'.

In general form, matrix is defined as a rectangular array of elements arranged in rows and columns as shown below:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

The matrix shown above has m rows and n columns. Matrix A, called matrix of order or dimension $m \times n$.

In A, an element a_{ij} , such as the first subscript refers to the number of row and the second to the number of columns to which the element belongs. For example, a_{13} belongs to the 1st row and 3rd column.

In general, a_{ij} refers to the element of jth row and ith column. If the matrix has total number of elements = m. n, arranged in m rows and n column, **it is said to be of the order of m by n**, which is often written as $m \times n$ matrix.

Remark 1:

- (i) A matrix is denoted by capital letters A, B, C, etc. of the English alphabets.
- (ii) First suffix of an element of the matrix indicates the position of row and second suffix of the element of the matrix indicates position of column.
e.g. a_{23} means it is an element in the second row and the third column.
- (iii) The order of a matrix is written as “number of rows $\times$ number of columns”

For example,

- (i) $A = \begin{bmatrix} 2 & 5 & 7 \\ 3 & 8 & 9 \end{bmatrix}$ is a matrix of order 2×3

(ii) $B = \begin{bmatrix} 9 & 6 \\ 1 & 0 \\ 8 & 4 \end{bmatrix}$ is a matrix of order 3×2

Let us consider some more examples:

Example 1: Write the order of the matrix

$$A = \begin{bmatrix} 9 & 7 & 8 & -3 & -8 \\ 4 & 3 & 6 & 1 & -10 \\ 10 & 12 & 15 & 2 & 5 \end{bmatrix}$$

Also write the elements $a_{23}, a_{14}, a_{35}, a_{22}, a_{31}, a_{32}$.

Solution: Order of the matrix A is 3×5 and the desired elements are:

$$a_{23} = 6, a_{14} = -3, a_{35} = 5, a_{22} = 3, a_{31} = 10, a_{32} = 12$$

6.3 TYPES OF MATRICES

On the basis of number of rows and number of columns and depending on the values of elements, the type of a matrix gets changed. Various types of matrices are explained as below:

A) Column Matrix and Row Matrices

Column Matrix

If the matrix consists of only one column as $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$, then it is called a 'column matrix' or 'column vector'.

For example, $\begin{bmatrix} 9 \\ 6 \\ 7 \end{bmatrix}, \begin{bmatrix} 9 \\ -3 \\ 2 \\ 8 \end{bmatrix}, \begin{bmatrix} 5 \\ -11 \end{bmatrix}$ are column matrices.

Remark 2: If a matrix has one element, e.g. $A = [6]$, then matrix A has only one row and only one column. So, it is both row matrix as well as column matrix.

Row Matrix

A matrix having only one row is called a row matrix.

For example, $[2 \ 5 \ 7], [8 \ 9], [1 \ 0 \ 3 \ 2]$ are row matrices.

B) Rectangular Matrix

A matrix of the order m by n is also called as 'rectangular matrix' if $m \neq n$. And if $m=n$, then the matrix is said to be a '**square matrix**' of order 'n'.

For example, $\begin{bmatrix} 2 & 5 & 7 \\ 3 & 8 & 9 \end{bmatrix}$ is a rectangular matrix having 2 rows and 3 columns.

C) Null Matrix

If all the elements of a matrix are zero, then it is called a 'null matrix' or 'zero matrix'. The null matrix of order $m \times n$ is denoted by $O_{m \times n}$.

$$\text{e.g., } 0_{2 \times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

If the elements of a matrix are not all zero, then it is called a 'non-zero matrix'. If the elements of a matrix are not all zero, then it is called a 'non-zero matrix'.

D) Negative of a Matrix

The *negative* of a matrix $A = [a_{ij}]_{m \times n}$ is $[-a_{ij}]_{m \times n}$ and is denoted by $(-A)$.

$$\text{Thus, if } A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix}$$

$$\text{then } -A = \begin{bmatrix} -a_1 & -b_1 & -c_1 \\ -a_2 & -b_2 & -c_2 \end{bmatrix}$$

For example,

$$\text{If } A = \begin{bmatrix} 10 & 5 & 5 \\ -4 & -8 & 6 \end{bmatrix}$$

$$\text{then } -A = \begin{bmatrix} -10 & -5 & -5 \\ 4 & 8 & -6 \end{bmatrix}$$

E) Diagonal Matrix

Principal Diagonal :

If $A = [a_{ij}]_{n \times n}$ is a square matrix, then the elements $a_{11}, a_{22}, \dots, a_{nn}$ constitute the principal diagonal of A .

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{n1} & a_{n1} & \dots & a_{nn} \end{bmatrix}$$

Principal diagonal

A square matrix $A = [a_{ij}]_{n \times n}$ is said to be diagonal matrix if $a_{ij} = 0, \forall i \neq j$.

For example,

$$(i) \text{ If } A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}, \text{ then it is a diagonal matrix because all its non-diagonal}$$

elements are zero. Sometimes, we denote it by writing $\text{diag. } [4, 3, 6]$.

$$(ii) A = \begin{bmatrix} 2 & 9 \\ 0 & 5 \end{bmatrix} \text{ is not a diagonal matrix because non-diagonal element } A_{12} \neq 0.$$

Remark 3:

- (i) For a diagonal matrix all non-diagonal elements must be zero.
- (ii) In a diagonal matrix some or all the diagonal elements may be zero.

Example 2: Write all the diagonal matrices of order 2×2 having its elements only 0 or 1.

Solution: For a diagonal matrix, all the non-diagonal elements are zero. Therefore, we are to write 0 and 1 in the diagonal elements in different ways, i.e. 0, 0; 0, 1; 1, 0; and 1, 1.

∴ possible diagonal matrices with elements only 0 and 1 are given below:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

F) Scalar Matrix

If the diagonal elements of a diagonal matrix be all equal, then the matrix is called a '**scalar matrix**'.

e. g., $\begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$ is a scalar matrix $[a]_{3 \times 3}$

Example of some more scalar matrices are: $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

G) Identity Matrix

If the diagonal elements of a diagonal matrix are all equal to 1, it is said to be a '**unit matrix**' or '**identity matrix**'. A unit matrix of order 'n' is denoted by I_n or simply by I. Thus,

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \dots & \dots & \dots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} = (a_{ij})_{n \times n} \text{ is a diagonal matrix if and only if}$$

$$a_{ij} = 0 \text{ for } i \neq j \text{ and } a_{ij} = 1 \text{ for } i = j$$

Example: $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is an identity matrix of order 3.

For example, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ all are identity (or Unit) matrices of

order 2 and 4 respectively.

H) Upper Triangular Matrix

A square matrix $A = [a_{ij}]_{n \times n}$ is said to be upper triangular matrix if all the elements below the principal diagonal are zero.

For example, $\begin{bmatrix} 2 & 5 \\ 0 & 7 \end{bmatrix}$, $\begin{bmatrix} 9 & 0 & 6 \\ 0 & 5 & 4 \\ 0 & 0 & 7 \end{bmatrix}$, $\begin{bmatrix} 8 & 0 & 5 & 0 \\ 0 & 9 & 3 & 0 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ all are upper triangular matrices.

But $\begin{bmatrix} 2 & 9 & 7 \\ 0 & 5 & 8 \\ 2 & 0 & 9 \end{bmatrix}$ is not an upper triangular matrix because one element below the principal diagonal, i.e. a_{31} is non zero, which is 2, in this case.

I) Lower Triangular Matrix

A square matrix $A = [a_{ij}]_{n \times n}$ is said to be lower triangular matrix if all the elements above the principal diagonal are zero.

For example, $\begin{bmatrix} 5 & 0 \\ 3 & 2 \end{bmatrix}$, $\begin{bmatrix} 3 & 0 & 0 \\ 2 & 6 & 0 \\ 1 & 9 & 7 \end{bmatrix}$, are lower triangular matrices of orders 2 and 3 respectively.

J) Comparable Matrices

Two matrices are said to be comparable if they are of the same order.

For example,

if $A = \begin{bmatrix} 2 & 5 & 3 \\ 6 & 8 & 9 \end{bmatrix}$, $B = \begin{bmatrix} a & b & c \\ x & y & z \end{bmatrix}$, then A and B are comparable because both are of the same order, i.e. of order 2×3 .

K) Equal Matrices

Two matrices are said to be equal if

- (i) they are of same order, and
- (ii) the corresponding elements of the matrices are equal.

For example, if $A = \begin{bmatrix} 2 & 8 \\ 3 & x \end{bmatrix}$, $B = \begin{bmatrix} a & 8 \\ 3 & 5 \end{bmatrix}$, then $A = B$, if $a = 2$, $x = 5$.

Check Your Progress – 1

1. Write all the possible orders of the matrix having 8 elements

.....

2. Construct $A = [a_{ij}]_{3 \times 2}$, where $a_{ij} = |i - j|$

.....

3. Write orders and types of the following matrices:

$$(i) \begin{bmatrix} 2 & 9 \\ 3 & 4 \end{bmatrix} \quad (ii) \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix} \quad (iii) \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix} \quad (iv) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (v) \begin{bmatrix} 2 & 5 & 7 \\ 0 & 8 & 0 \\ 0 & 0 & 9 \end{bmatrix} \quad (vi) \\ \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 7 & 6 \end{bmatrix} \quad (vii) \begin{bmatrix} 2 \\ 9 \\ 6 \end{bmatrix} \quad (viii) [8 \quad 9 \quad 1 \quad 5] \quad (ix) \begin{bmatrix} 2 & 9 & 3 \\ 6 & 4 & 5 \end{bmatrix}$$

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4.

(i) If $\begin{bmatrix} 3 & x+y \\ xy & 7+z \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 8 & 4 \end{bmatrix}$, find x, y, z.

(ii) If $\begin{bmatrix} a+5 & -2a & b+6 \\ 2c & 3b+2 & x \\ y+1 & z+3 & x+2 \end{bmatrix} = \begin{bmatrix} 2 & a+9 & 11 \\ c+4 & -b+22 & 3-x \\ 2 & 2z+3 & 5-x \end{bmatrix}$

find a, b, c, x, y, z.

.....
.....
.....

5. Find the values of x, y, z, w if $\begin{bmatrix} 3x-2y & z+w \\ 3z-w & x+y \end{bmatrix} = \begin{bmatrix} -1 & 7 \\ 5 & 3 \end{bmatrix}$.

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6.4 OPERATIONS ON MATRICES

From our school times, we have learned several concepts related to basic mathematics. First you learn the natural numbers and then learns how these numbers are added, subtracted, multiplied and divided. Similarly, here also we now see as to how such operations (**except division**) are applied on matrices.

These operations are explained by first giving a general formula and then examples.

Note: Division of a matrix by another matrix is meaning less and hence it is not permitted in case of matrices.

6.4.1 Addition and Subtraction of Matrices

If the matrices A and B are of the same order (m x n), then they can be added or subtracted.

If $A=[a_{ij}]$ and $B=[b_{ij}]$, then $A+B = [a_{ij} + b_{ij}]$ and $A-B = [a_{ij}-b_{ij}]$

In other words, corresponding elements are added or subtracted. The new matrix (the resultant) will be of the same order (m x n).

Example: $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1+0 & 2+1 & 3+2 \\ 3+2 & 4+3 & 5+4 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 7 & 9 \end{bmatrix}$

Properties of Addition of Matrices

- i) $A+B=B+A$ (so, matrix addition is commutative)
- ii) $(A+B)+C=A+(B+C)$ where C is the third matrix (so, matrix addition is associative).
- iii) $A + \mathbf{O} = \mathbf{O} + A = A$, where **O denotes zero matrix of the same order.**
Thus, **zero matrix plays the same role in matrix algebra as zero in ordinary algebra.**

For a given matrix A, there exists a matrix B of the same order such that $A + B = \mathbf{O} = B + A$.

Here B is called additive inverse of A. (existence of additive inverse)

Note: If A is a matrix and k is a scalar, then kA is a new matrix formed by multiplying **every single entry** of A by the value k.

If $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$ and $k = 7$, then $kA = 7A = \begin{bmatrix} 7 \times 3 & 7 \times 4 \\ 7 \times 5 & 7 \times 6 \end{bmatrix} = \begin{bmatrix} 21 & 28 \\ 35 & 42 \end{bmatrix}$.

Example 3:

- (i) If $A = \begin{bmatrix} 2 & 3 & 4 \\ 7 & 5 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 5 & 6 \\ 2 & 9 & 8 \end{bmatrix}$ then find $A+B$
- ii) If $A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 6 & 9 \\ 2 & 4 & 5 \end{bmatrix}$ then find $A + B$

Solution

- (i) $A + B = \begin{bmatrix} 2+1 & 3+5 & 4+6 \\ 7+2 & 5+9 & 1+8 \end{bmatrix} = \begin{bmatrix} 3 & 8 & 10 \\ 9 & 14 & 9 \end{bmatrix}$.
- (ii) Here $A+B$ does not make any sense because A and B are of different orders.

Subtraction of Matrices

Subtraction of two matrices A and B make sense only if they are of the same order, and is given by

$A-B = A + (-B) = A + (-1)B$, i.e. $A - B$ means addition of two matrices A and $-B$. So, if $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n}$, then $A-B = [a_{ij} + (-1)b_{ij}]_{m \times n} = [a_{ij} - b_{ij}]_{m \times n}$

If $A = \begin{bmatrix} 2 & 3 & 4 \\ 7 & 5 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 5 & 6 \\ 2 & 9 & 8 \end{bmatrix}$

then $A-B = \begin{bmatrix} 2-1 & 3-5 & 4-6 \\ 7-2 & 5-9 & 1-8 \end{bmatrix}$

$$= \begin{bmatrix} 1 & -2 & -2 \\ 5 & -4 & -7 \end{bmatrix}$$

6.4.2 Matrix Multiplication

If the number of columns of a matrix A be equal to the number of rows of another matrix B, then the matrices A and B are said to be '**conformable for the product**' AB, and the product AB is said to be defined.

If $A = [a_{ij}]_{m \times n}$ and $B = [b_{jk}]_{n \times p}$

Then $AB = [c_{ik}]_{m \times p}$ where $c_{ik} = \sum_{j=1}^n a_{ij} b_{jk}$

$$\text{Thus, } \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 5 & 2 \\ 9 & 7 \end{bmatrix} = \begin{bmatrix} 1+10+27 & 3+4+21 \\ 0+5+18 & 0+2+14 \end{bmatrix} = \begin{bmatrix} 38 & 28 \\ 23 & 16 \end{bmatrix}$$

Further if $AB = C = [c_{ij}]^{m \times p}$ where $c_{ij} = (i, j)^{th}$ element of C and is equal to:

$= \sum_{k=1}^n a_{ik} b_{kj}$, i.e. sum of product of first, second, third, ... elements of i^{th} row of A with first, second, third, ... , elements of j^{th} column of B respectively.

You may notice that the number of rows in AB = number of rows in A, and number of columns in AB = number of columns in B.

Properties of Matrix Multiplication:

If A, B, C and D are four matrices such that corresponding multiplications hold then

- i) $(AB)C = A(BC)$ [matrix multiplication is associative.]
- ii) $(B+C)D = BD + CD$ [so, matrix multiplication is distributive with respect to addition]
 - (a) $A(B + C) = AB + AC$ (left distributive law)
 - (b) $(A + B)C = AC + BC$ (right distributive law)

iii) If A is a square matrix of order n, then

$$I_n A = A I_n = A, \text{ where } I_n \text{ is the identity matrix of order } n.$$

Remark 4: Commutative law does not hold, in general,

i.e. $AB \neq BA$, in general. But for some cases AB may be equal to BA. This has been explained below:

(i) AB may be defined but BA may not be defined and hence $AB \neq BA$ in this case.

For example, let A be a matrix of order 3×2 and B be a matrix of order 2×4 . Here AB is defined and is of order 3×4 .

But BA is not defined ($\because$ number of columns of B $\neq$ number of rows of A).

(ii) AB and BA both may be defined but may not be of same order and hence $AB \neq BA$.

For example, let A be a matrix of order 3×2 and B be a matrix of order 2×3 .

Here as number of columns of A = number of rows of B.

$\therefore$ AB is defined and is of order 3×3 .

Also, number of columns of B = number of rows of A.

Hence BA is defined but of order 2×2 .

$\therefore AB \neq BA$.

(iii) AB and BA both may be defined and of same order but even then, they may not be equal.

Let

$$A = \begin{bmatrix} 5 & 4 \\ 2 & 7 \end{bmatrix}, B = \begin{bmatrix} 6 & 8 \\ 2 & 9 \end{bmatrix}$$

Here, AB and BA both are defined and are of same order.

$$AB = \begin{bmatrix} 5 & 4 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} 6 & 8 \\ 2 & 9 \end{bmatrix} = \begin{bmatrix} 30 + 8 & 40 + 36 \\ 12 + 14 & 16 + 63 \end{bmatrix} = \begin{bmatrix} 38 & 76 \\ 26 & 79 \end{bmatrix} \text{ and}$$

$$BA = \begin{bmatrix} 6 & 8 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 2 & 7 \end{bmatrix} = \begin{bmatrix} 30 + 16 & 24 + 56 \\ 10 + 18 & 8 + 63 \end{bmatrix} = \begin{bmatrix} 46 & 80 \\ 28 & 71 \end{bmatrix}.$$

So, $AB \neq BA$.

However, sometimes, we may observe that $AB = BA$.

For example,

Let

$$A = \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix}, \text{ and } B = \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix}.$$

$$\text{Here } AB = \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} 10 + 12 & 6 - 6 \\ 20 - 20 & 12 + 10 \end{bmatrix} = \begin{bmatrix} 22 & 0 \\ 0 & 22 \end{bmatrix} \text{ and}$$

$$BA = \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 10 + 12 & -15 + 15 \\ -8 + 8 & 12 + 10 \end{bmatrix} = \begin{bmatrix} 22 & 0 \\ 0 & 22 \end{bmatrix}.$$

Here, $AB = BA$.

Example 4: If $A = \begin{bmatrix} 2 & 4 & 5 \\ -3 & 6 & 7 \\ 1 & 8 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 6 & 2 \\ 1 & 4 & 5 \\ 8 & 7 & -1 \end{bmatrix}$, then evaluate the

followings:

- (i) $3A + 2B$
- (ii) $2A - 3B$
- (iii) AB

Solution:

(i) $3A + 2B =$

$$3 \begin{bmatrix} 2 & 4 & 5 \\ -3 & 6 & 7 \\ 1 & 8 & 9 \end{bmatrix} + 2 \begin{bmatrix} 3 & 6 & 2 \\ 1 & 4 & 5 \\ 8 & 7 & -1 \end{bmatrix} = \begin{bmatrix} 6 & 12 & 15 \\ -9 & 18 & 21 \\ 3 & 24 & 27 \end{bmatrix} + \begin{bmatrix} 6 & 12 & 4 \\ 2 & 8 & 10 \\ 16 & 14 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 6+6 & 12+12 & 15+4 \\ -9+2 & 18+8 & 21+10 \\ 3+16 & 24+14 & 27-2 \end{bmatrix} = \begin{bmatrix} 12 & 24 & 19 \\ -7 & 26 & 31 \\ 19 & 38 & 25 \end{bmatrix}$$

$$(i) \quad 2A - 3B = 2 \begin{bmatrix} 2 & 4 & 5 \\ -3 & 6 & 7 \\ 1 & 8 & 9 \end{bmatrix} - 3 \begin{bmatrix} 3 & 6 & 2 \\ 1 & 4 & 5 \\ 8 & 7 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 10 \\ -6 & 12 & 14 \\ 2 & 16 & 18 \end{bmatrix} -$$

$$\begin{bmatrix} 9 & 18 & 6 \\ 3 & 12 & 15 \\ 24 & 21 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 4-9 & 8-18 & 10-6 \\ -6-3 & 12-12 & 14-15 \\ 2-24 & 16-21 & 18+3 \end{bmatrix} = \begin{bmatrix} -5 & -10 & 4 \\ -9 & 0 & -1 \\ -22 & -5 & 21 \end{bmatrix}$$

$$(iii) \quad AB = \begin{bmatrix} 2 & 4 & 5 \\ -3 & 6 & 7 \\ 1 & 8 & 9 \end{bmatrix} \begin{bmatrix} 3 & 6 & 2 \\ 1 & 4 & 5 \\ 8 & 7 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 6+4+40 & 12+16+35 & 4+20-5 \\ -9+6+56 & -18+24+49 & -6+30-7 \\ 3+8+72 & 6+32+63 & 2+40-9 \end{bmatrix}$$

$$= \begin{bmatrix} 50 & 63 & 19 \\ 53 & 55 & 17 \\ 83 & 101 & 33 \end{bmatrix}$$

6.4.3 Transpose of a matrix

If $A = [a_{ij}]_{m \times n}$ a matrix, the transpose of A denoted by A' or A^T is defined to be the matrix $[b_{ij}]_{n \times m}$ where $[b_{ij}] = [a_{ji}] \forall i \text{ and } j$. In other words, transpose of a matrix A is denoted by A' or A^T and is obtained by interchanging rows and columns of A .

For example, if $A = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 8 \end{bmatrix}$ then $A^T = \begin{bmatrix} 2 & 5 \\ 3 & 6 \\ 4 & 8 \end{bmatrix}$.

Note: $(A^T)^T = A$.

Properties of Transpose Matrices

$$(i) \quad (A')' = A$$

$$(ii) \quad (kA)' = kA', \quad \text{where } k \text{ is a scalar}$$

$$(iii) \quad (A + B)' = A' + B'$$

$$(iv) \quad (A - B)' = A' - B'$$

$$(v) \quad (AB)' = B'A'$$

Example 5:

$$A = \begin{bmatrix} 2 & -1 \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 6 & 8 \\ 10 & -12 \end{bmatrix}, C = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} \text{ then}$$

$$[A - B + C] = \begin{bmatrix} -4 & -10 \\ -10 & 11 \end{bmatrix}$$

or alternatively

$$A^T - B^T + C^T = \begin{bmatrix} 2 & 0 \\ -1 & -1 \end{bmatrix} - \begin{bmatrix} 6 & 10 \\ 8 & -12 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} -4 & -10 \\ -10 & 11 \end{bmatrix}$$

Example 6: $A = \begin{bmatrix} 3 & 0 \\ -4 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 5 & -7 \\ 0 & -1 & 8 \end{bmatrix}$, $C = \begin{bmatrix} 6 \\ -1 \\ 0 \end{bmatrix}$ then

$$ABC = \begin{bmatrix} 39 \\ -53 \end{bmatrix} [ABC]^T = [39 \ -53] =$$

$$C^T B^T A^T = [6 \ -1 \ 0] \begin{bmatrix} 3 & 0 \\ 5 & -1 \\ -7 & 8 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 0 & -1 \end{bmatrix}$$

6.4.4 Integral Powers of a Square Matrix

Here, we will learn how higher powers of a square matrix A are evaluated. We define

$$A^2 = A.A$$

$$A^3 = A^2 A \quad \text{or} \quad A^3 = A A^2$$

$$A^4 = A^3 A \quad \text{or} \quad A^4 = A A^3 \quad \text{or} \quad A^4 = A^2 A^2$$

and so on

in general, $A^{p+q} = A^p A^q = A^q A^p$.

Remark 5:

$$(i) \quad (A + B)^2 = A^2 + AB + BA + B^2.$$

$$(ii) \quad (A + B)^2 = A^2 + 2AB + B^2 \quad \text{if and only if} \quad AB = BA.$$

Example 7: If $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ then find A^4 .

Solution:

$$A^2 = AA = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 1+8 & 2+6 \\ 4+12 & 8+9 \end{bmatrix} = \begin{bmatrix} 9 & 8 \\ 16 & 17 \end{bmatrix}$$

$$A^4 = A^2 A^2 = \begin{bmatrix} 9 & 8 \\ 16 & 17 \end{bmatrix} \begin{bmatrix} 9 & 8 \\ 16 & 17 \end{bmatrix} = \begin{bmatrix} 81+128 & 72+136 \\ 144+272 & 128+289 \end{bmatrix} \\ = \begin{bmatrix} 209 & 208 \\ 416 & 417 \end{bmatrix}$$

6.4.5 Adjoint and Reciprocal Matrices

Cofactor: Cofactor of a square matrix Let $A = [a_{ij}]_{n \times n}$ be a square matrix, the cofactor of element of a_{ij} is $(-1)^{i+j} M_{ij}$ where M_{ij} is determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting i^{th} row and j^{th} column of A .

If $A = [a_{ij}]_{n \times n}$ be a square matrix, then the transpose of the matrix $[A_{ij}]_{n \times n}$ Whose elements are the cofactors of the corresponding element in $|A|$ is called the 'adjoint' or 'adjugate' matrix of A and is denoted by **adjA**. It is equal to $[A_{ij}]_{n \times n}$.

In section 6.4.8 we will discuss inverse of a square matrix. But in order define

inverse of a square matrix we use the concept of adjoint of the square matrix, so in this section we are going to discuss adjoint of the square matrix.

Adjoint of a Matrix: Let $A = [a_{ij}]_{n \times n}$ be a square matrix of order $n \times n$, then adjoint of A is denoted by $\text{adj}A$ and is defined as

$$\text{adj}A = \begin{bmatrix} A_{11} & A_{12} & A_{13} & \dots & A_{1n} \\ A_{21} & A_{22} & A_{23} & \dots & A_{2n} \\ A_{31} & A_{32} & A_{33} & \dots & A_{3n} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ A_{n1} & A_{n2} & A_{n3} & \dots & A_{nn} \end{bmatrix}^T$$

where A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A .

It may be verified that $A(\text{adj}A) = (\text{adj}A)A = |A|I$,

where I is an identity matrix of order n .

Example 8: Find the $\text{adj}A$, where $A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 0 \\ -1 & -2 & 1 \end{bmatrix}$.

Also verify that $A(\text{adj}A) = (\text{adj}A)A = |A|I$.

Solution: $A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 0 \\ -1 & -2 & 1 \end{bmatrix}$

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A .

$$\therefore A_{11} = (-1)^{1+1} \begin{vmatrix} 5 & 0 \\ -2 & 1 \end{vmatrix} = (-1)^2 (5 - 0) = 5$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} -4 & 0 \\ -1 & 1 \end{vmatrix} = (-1)^3 (-4 - 0) = 4$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} -4 & 5 \\ -1 & -2 \end{vmatrix} = (-1)^4 (8 + 5) = 13$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 2 & 3 \\ -2 & 1 \end{vmatrix} = (-1)^3 (2 + 6) = -8$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = (-1)^4 (1 + 3) = 4$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ -1 & -2 \end{vmatrix} = (-1)^5 (-2 + 2) = 0$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} 2 & 3 \\ 5 & 0 \end{vmatrix} = (-1)^4 (0 - 15) = -15$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 3 \\ -4 & 0 \end{vmatrix} = (-1)^5(0+12) = -12$$

$$A_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 2 \\ -4 & 5 \end{vmatrix} = (-1)^6(5+8) = 13$$

$$\therefore \text{adj}A = \begin{bmatrix} 5 & 4 & 13 \\ -8 & 4 & 0 \\ -15 & -12 & 13 \end{bmatrix}^T = \begin{bmatrix} 5 & -8 & -15 \\ 4 & 4 & -12 \\ 13 & 0 & 13 \end{bmatrix}$$

To verify $A(\text{adj}A) = (\text{adj}A)A = |A|I$, we proceed as follows

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & 0 \\ -1 & -2 & 1 \end{vmatrix}$$

Expanding along C_3

$$|A| = 3(8+5) - 0 + 1(5+8) = 39 + 13 = 52 \quad \dots (1)$$

$$\begin{aligned} A(\text{adj}A) &= \begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 0 \\ -1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 5 & -8 & -15 \\ 4 & 4 & -12 \\ 13 & 0 & 13 \end{bmatrix} \\ &= \begin{bmatrix} 5+8+39 & -8+8+0 & -15-24+39 \\ -20+20+0 & 32+20+0 & 60-60+0 \\ -5-8+13 & 8-8+0 & 15+24+13 \end{bmatrix} \\ &= \begin{bmatrix} 52 & 0 & 0 \\ 0 & 52 & 0 \\ 0 & 0 & 52 \end{bmatrix} \\ &= 52 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 52I = |A|I \end{aligned}$$

... (2) [Using (1)]

$$\begin{aligned} (\text{adj}A)A &= \begin{bmatrix} 5 & -8 & -15 \\ 4 & 4 & -12 \\ 13 & 0 & 13 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 0 \\ -1 & -2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 5+32+15 & 10-40+30 & 15-0-15 \\ 4-16+12 & 8+20+24 & 12+0-12 \\ 13+0-13 & 26+0-26 & 39+0+13 \end{bmatrix} = \begin{bmatrix} 52 & 0 & 0 \\ 0 & 52 & 0 \\ 0 & 0 & 52 \end{bmatrix} \\ &= 52 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 52I = |A|I \end{aligned} \quad \dots (3)$$

From (2) and (3), we have $A(\text{adj}A) = (\text{adj}A)A = |A|I$

6.4.6 Trace of a Matrix

Trace of a square matrix $A = [a_{ij}]_{n \times n}$ is denoted by $\text{tr}(A)$ and is defined as
 $\text{tr}(A) = \text{sum of the elements of principal diagonal of the matrix.}$

$$\text{i.e. } \text{tr}(A) = a_{11} + a_{22} + a_{33} + \dots + a_{nn}$$

Thus, in the matrix $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & 1 \\ 3 & 3 & -1 \end{bmatrix}$ the trace (tr) is $= 1 + 0 - 1 = 0$

For example, if $A = \begin{bmatrix} 2 & 3 & 5 \\ 6 & 8 & 4 \\ 9 & 1 & -3 \end{bmatrix}$ then $\text{tr}(A) = 2 + 8 + (-3) = 7$.

Properties:

If $A = [a_{ij}]_{n \times n}$ and $B = [b_{ij}]_{n \times n}$ then

- i) $\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$
- ii) $\text{tr}(A^T) = \text{tr}(A)$
- iii) $\text{tr}(AB) = \text{tr}(BA)$
- iv) $\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB)$
- v) $\text{tr}(kA) = k \text{tr}(A)$, where k is a scalar

Remark 6: $\text{tr}(AB) \neq \text{tr}(A) \text{tr}(B)$

Here is an exercise for you.

6.4.7 Sub Matrices and Minors

Any matrix obtained by omitting some rows and some columns of a given $m \times n$ matrix A is called 'sub matrix' of A

Thus, $\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$ is a sub matrix of $\begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{bmatrix}$

The determinant of the square matrix of order ' r ', obtained from a given $m \times n$ matrix A by omitting $(m - r)$ rows and $(n - r)$ columns, is called a '**minor**' of A of order r '.

Thus, $\begin{bmatrix} a & h \\ h & b \end{bmatrix}$ is a minor of $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$ of order 2

6.4.8 Inverse of a Matrix

You are familiar with the concept of the multiplicative inverse of a real number.

e.g. you know the multiplicative inverse of 5, $\frac{2}{3}$, etc.

Multiplicative inverse of 5 is $\frac{1}{5}$ and of $\frac{2}{3}$ is $\frac{3}{2}$, etc.

i.e. when a number is multiplied with its multiplicative inverse, it gives 1.

Similarly, in case of matrices, when a square matrix is multiplied with its multiplicative inverse, it gives identity matrix (I). That is, in case of matrices, identity, matrix plays the role as 1 plays in case of real numbers.

Inverse of a Matrix: Let A be a square matrix of order n. If there exists a square matrix B of order n such that:

$$AB = BA = I_n, \text{ then B is known as inverse of A and is denoted by } A^{-1}.$$

$$\text{As verified earlier that } A(\text{adj}A) = (\text{adj}A)A = |A|I,$$

$$\therefore A(\text{adj}A) = |A|I$$

$$\Rightarrow A \left[\frac{1}{|A|} (\text{adj}A) \right] = I, \text{ provided } |A| \neq 0$$

Now, by definition of inverse of a matrix, $\frac{1}{|A|} (\text{adj}A)$ is inverse of A.

$$\text{i.e. } A^{-1} = \frac{1}{|A|} (\text{adj}A)$$

Definition: Inverse/reciprocal of a non-singular n x n square matrix A is a matrix B of the same order as that A, which satisfies the matrix equation $AB = BA = I$, where I is the n x n unit matrix. Inverse/reciprocal of matrix A is denoted as A^{-1} .

It can be shown that the inverse of a non-singular matrix A is given by

$$A^{-1} = \frac{1}{|A|} (\text{Adj. } A)$$

Properties of Inverse of a Matrix

- i) The inverse of a square matrix, if exists is unique.
- ii) A necessary and sufficient condition that the inverse of a square matrix A to exist, is that A is non-singular.
- iii) A^{-1} exists if and only if $|A| \neq 0$.
- iv) $(AB)^{-1} = B^{-1}A^{-1}$
- v) $[AB]^{-1} = B^{-1}.A^{-1}$ (this is known as 'law a/reversal a/inverse').
- vi) $(A^{-1})^{-1} = A$
- vii) $(A^{-1})^T = (A^T)^{-1}$ and $A^T = I = A^T (A^{-1})^T$ (operations of reciprocating and transposing are commutative)

Note:

1. A square matrix $A_{n \times n}$ with $|A| \neq 0$ is called a **non-singular** matrix.
2. $A_{n \times n}$ is called **singular** matrix if $|A| = 0$.

To find inverse of a square matrix use following steps:

If $|A| = 0$, A^{-1} does not exist.

Note: If $|A| \neq 0$ then find $\text{adj}A$ and $A^{-1} = \frac{1}{|A|} (\text{adj}A)$.

Example 9: Find A^{-1} , where $A = \begin{bmatrix} 4 & 2 & 3 \\ 0 & -1 & -2 \\ 5 & -3 & 6 \end{bmatrix}$

Solution: $A = \begin{bmatrix} 4 & 2 & 3 \\ 0 & -1 & -2 \\ 5 & -3 & 6 \end{bmatrix}$

$$|A| = 4(-6 - 6) - 2(0 + 10) + 3(0 + 5)$$

$$= -48 - 20 + 15 = -53 \neq 0$$

$\therefore A^{-1}$ exists.

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A.

$$\therefore A_{11} = (-1)^{1+1}(-6 - 6) = -12, \quad A_{12} = (-1)^{1+2}(0 + 10) = -10$$

$$A_{13} = (-1)^{1+3}(0 + 5) = 5, \quad A_{21} = (-1)^{2+1}(12 + 9) = -21$$

$$A_{22} = (-1)^{2+2}(24 - 15) = 9, \quad A_{23} = (-1)^{2+3}(-12 - 10) = 22$$

$$A_{31} = (-1)^{3+1}(-4 + 3) = -1, \quad A_{32} = (-1)^{3+2}(-8 - 0) = 8$$

$$A_{33} = (-1)^{3+3}(-4 - 0) = -4$$

$$\therefore \text{adj}A = \begin{bmatrix} -12 & -10 & 5 \\ -21 & 9 & 22 \\ -1 & 8 & -4 \end{bmatrix}^T = \begin{bmatrix} -12 & -21 & -1 \\ -10 & 9 & 8 \\ 5 & 22 & -4 \end{bmatrix}$$

$$\text{and } A^{-1} = \frac{1}{|A|}(\text{adj}A) = \frac{1}{-53} \begin{bmatrix} -12 & -21 & -1 \\ -10 & 9 & 8 \\ 5 & 22 & -4 \end{bmatrix}$$

Example 10: $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{bmatrix} \therefore |A| = 1(1 - 6) -$

$$1(-2 - 9) + 1(4 + 3)$$

$$= -5 + 11 + 7 = 13 \text{ i.e., } |A| \neq 0$$

So A is a non-singular matrix, its inverse exists.

Now, matrix of cofactors of $A = \begin{bmatrix} -5 & 11 & 7 \\ 3 & -4 & 1 \\ 4 & -2 & -3 \end{bmatrix}$

$$\therefore \text{Adj. } A = \begin{bmatrix} -5 & 3 & 4 \\ 11 & -4 & 1 \\ 7 & 1 & -3 \end{bmatrix} \text{ and } A^{-1} = \frac{1}{13} \begin{bmatrix} -5 & 3 & 4 \\ 11 & -4 & 1 \\ 7 & 1 & -3 \end{bmatrix}$$

Check Your Progress -2

1. If $3X + 2Y = \begin{bmatrix} 4 & 13 \\ 18 & 13 \end{bmatrix}$ and $2X - 3Y = \begin{bmatrix} 7 & 0 \\ -1 & -13 \end{bmatrix}$, then find matrices X and Y.

.....

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.....

2. Find AB, if defined, in each of the following cases:

(i) $A = \begin{bmatrix} 5 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ (ii) $A = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 6 \end{bmatrix}$

(iii) $A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & 5 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$

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3. Evaluate the product $\begin{bmatrix} 2 & 3 & 5 \end{bmatrix} \begin{bmatrix} 4 & 5 \\ 0 & 1 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 \\ 5 & 6 & 8 \end{bmatrix}$.

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4. If $A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$, then find A^8 .

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5. If $A = \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix}$ then verify that $A (\text{adj} A) = (\text{adj} A) A = |A| I_2$.

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6. (i) Find trace of the matrix A, where $A = \begin{bmatrix} 8 & 7 \\ 5 & 6 \end{bmatrix}$

- (ii) Find trace of the matrices I_2, I_3, I_n .

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7. For $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix}$, verify the result $(AB)^{-1} = B^{-1} A^{-1}$.

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8. For $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix}$, verify the result $(AB)^{-1} = B^{-1}A^{-1}$.

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6.5 SOME OTHER TYPES OF MATRICES

6.5.1 Orthogonal Matrix

A square matrix A is said to be orthogonal, if its transpose A^T produces a unit matrix when multiplied by A , i.e., $A^T A = I$.

Some properties of orthogonal matrices:

- i) Orthogonal matrices are non-singular, i.e., if $A^T A = I$ then $|A| \neq 0$
- ii) Unit matrices are orthogonal, i.e.,
 $I^T I = II = I$
- iii) The value of the determinant of an orthogonal matrix is either 1 or (-1).
If A is an orthogonal matrix, then $A^T A = I$
 $\therefore |A^T A| = |A^T| |A| = |A|^2 = |I|$
 $\therefore |A| = \pm 1$
- iv) The product of two orthogonal matrices is orthogonal.
- v) The transpose of an orthogonal matrix is orthogonal.
- vi) The inverse of an orthogonal matrix is orthogonal.

6.5.2 Symmetric and Skew-Symmetric Matrices

A square matrix A is said to be 'symmetric' if its transpose be equal to the matrix itself, i.e., if $A^T = A$. Hence, the matrix $[a_{ij}]$ is symmetric, if $a_{ij} = a_{ji}$ for all i and j . For example,

$\begin{bmatrix} a & b & c \\ b & e & d \\ c & d & f \end{bmatrix}$ is a symmetric matrix.

A square matrix A is said to be 'skew-symmetric', if its transpose is equal to the negative of the matrix itself. i.e., if $A^T = -A$. Hence, the matrix $[a_{ij}]$ is skew-symmetric, if $a_{ij} = -a_{ji}$ for all i and j . The diagonal elements of skew-symmetric matrix are obviously zero. For example,

$\begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$ is a skew – symmetric matrix.

Some Properties:

- i) The product of a matrix and its transpose is a symmetric matrix

$$\text{Now, } [A A^T]^T = [A^T]^T [A]^T = A[A]^T$$

Hence, AA^T is symmetric.

- ii) The product of two symmetric matrices of the same order is symmetric if the product is commutative.

If A and B be two symmetric matrices of the same order and $AB = BA$, then

$$[AB]^T = B^T A^T = BA = AB$$

Hence, AB is symmetric.

- iii) If A is a square matrix, then $(A + A^T)$ is symmetric and $(A - A^T)$ is skew-symmetric.

$$(A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$$

Thus, $(A + A^T)$ is symmetric.

$$\text{Similarly, } (A - A^T)^T = A^T - (A^T)^T = A^T - A = -(A - A^T)$$

Hence, $(A - A^T)$ is skew-symmetric.

- iv) Every square matrix can be expressed as a sum of a symmetric matrix and a skew symmetric matrix uniquely.

Let A be a square matrix, then,

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

Example 11: If $A = \begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix}$ then show that

- (i) $\frac{1}{2}(A + A^T)$ is symmetric, and (ii) $\frac{1}{2}(A - A^T)$ is skew-symmetric.

Solution:

$$(i) \text{ Let } P = \frac{1}{2}(A + A^T) = \frac{1}{2} \left(\begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix} + \begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix}^T \right)$$

$$= \frac{1}{2} \left(\begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix} + \begin{bmatrix} 3 & -2 \\ 5 & 4 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} 6 & 3 \\ 3 & 8 \end{bmatrix} = \begin{bmatrix} 3 & 3/2 \\ 3/2 & 4 \end{bmatrix} \quad \dots (1)$$

$$P^T = \begin{bmatrix} 3 & 3/2 \\ 3/2 & 4 \end{bmatrix}^T = \begin{bmatrix} 3 & 3/2 \\ 3/2 & 4 \end{bmatrix} \quad \dots (2)$$

From (1) and (2)

$$P^T = P \Rightarrow P \text{ is symmetric, i.e. } \frac{1}{2}(A + A^T) \text{ is symmetric.}$$

- (ii) Let Q =

$$\begin{aligned} \frac{1}{2}(A - A^T) &= \frac{1}{2} \left(\begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix} - \begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix}^T \right) = \frac{1}{2} \left(\begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix} - \begin{bmatrix} 3 & -2 \\ 5 & 4 \end{bmatrix} \right) \\ &= \frac{1}{2} \begin{bmatrix} 3-3 & 5+2 \\ -2-5 & 4-4 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & 7 \\ -7 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 7/2 \\ -7/2 & 0 \end{bmatrix} \end{aligned}$$

$$Q^T = \begin{bmatrix} 0 & 7/2 \\ -7/2 & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & -7/2 \\ 7/2 & 0 \end{bmatrix} = - \begin{bmatrix} 0 & 7/2 \\ -7/2 & 0 \end{bmatrix} = -Q$$

$\Rightarrow Q$ is skew symmetric, i.e. $\frac{1}{2}(A - A^T)$ is skew symmetric.

Remark 7:

$$A = \frac{1}{2}A + \frac{1}{2}A = \frac{1}{2}A + \frac{1}{2}A^T + \frac{1}{2}A - \frac{1}{2}A^T = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = P + Q$$

i.e. $A = P + Q$, where P is symmetric and Q is skew symmetric.

i.e. every square matrix can be expressed as a sum of a symmetric and a skew-symmetric matrix

6.5.3 Idempotent Matrix

An idempotent matrix is a **square matrix** that, when multiplied by itself, results in the original matrix.

In simple terms, if A is an idempotent matrix, then the following equation holds true:

$$A^2 = A$$

These matrices are used in Linear Algebra and Geometry. In linear algebra, an idempotent matrix represents a projection operator. When you apply the transformation twice, you get the same result as applying it once. Also, idempotent matrices are used in Computer Graphics, Network Analysis etc.

6.5.4 Partitioned Matrices

It is often convenient to partition a matrix into sub matrices. These sub matrices can then be treated as scalars (single values) in performing operations on the original matrix. Partitioning of a matrix is indicated by horizontal and vertical lines between rows and columns.

The $m \times n$ matrix A may be partitioned as follows $A = (A_1 : A_2)$

If they are partitioned conformably, partitioned matrices can be added, subtracted or multiplied. If an $m \times n$ matrix A is partitioned $A = (A_1 : A_2)$, where A_1 is $m \times n_1$, A_2 is $m \times n_2$ and $n_1 + n_2 = n$, and $m \times n$ matrix B is partitioned $B = (B_1 : B_2)$, where B_1 is $m \times n_1$, B_2 is $m \times n_2$ and $n_1 + n_2 = n$, then $A \pm B = [A_1 \pm B_1 : A_2 \pm B_2]$

Similarly, if $A = \begin{bmatrix} A_1 \\ \dots \\ A_2 \end{bmatrix}$, where A is $m \times n$, A_1 is $m_1 \times n$, A_2 is $m_2 \times n$ and

$m_1 + m_2 = m$, and $B = \begin{bmatrix} B_1 \\ \dots \\ B_2 \end{bmatrix}$, where B is $m \times n$, B_1 is $m_1 \times n$, B_2 is $m_2 \times n$ and

$m_1 + m_2 = m$ then $A \pm B = \begin{bmatrix} A_1 \pm B_1 \\ \dots \\ A_2 \pm B_2 \end{bmatrix}$

Partitioning is frequently conceptually convenient when matrices are to be added or subtracted, but the computational advantages of matrix partitioning

are associated primarily with multiplication and other more complex operations. Partitioned matrices must be partitioned conformable for multiplication. If matrix A of order $m \times n$ is partitioned $A = (A_1 : A_2)$, where A_1 is $m \times n_1$, A_2 is $m \times n_2$ and $n_1 + n_2 = n$ and $n \times p$ matrix B is partitioned

$$B = \begin{bmatrix} B_1 \\ \dots \\ B_2 \end{bmatrix}, \text{ where } B_1 \text{ is } n_1 \times p \text{ and } B_2 \text{ is } n_2 \times p \text{ the } AB = [A_1 : A_2] \begin{bmatrix} B_1 \\ \dots \\ B_2 \end{bmatrix} = A_1 B_1 + A_2 B_2.$$

Example 12: if $A = \begin{bmatrix} A_{11} & \vdots & A_{12} \\ \dots & \dots & \dots \\ A_{21} & \vdots & A_{22} \end{bmatrix}$ where A_{11} is $m_1 \times n_1$, A_{21} is

$m_2 \times n_1$, A_{12} is $m_1 \times n_2$, A_{22} is $m_2 \times n_2$ and $m_1 + m_2 = m$, $n_1 + n_2 = n$ and if

$$B = \begin{bmatrix} B_{11} & \vdots & B_{12} \\ \dots & \dots & \dots \\ B_{21} & \vdots & B_{22} \end{bmatrix}, \text{ where } B_{11} \text{ is } n_1 \times p_1, B_{12} \text{ is } n_1 \times p_2, B_{21} \text{ is } n_2 \times p_1, B_{22} \text{ is } n_2 \times p_2$$

and $n_1 + n_2 = n$, $p_1 + p_2 = p$, then A and B are partitioned conformably for

$$\begin{aligned} \text{multiplication and } AB &= \begin{bmatrix} A_{11} & \vdots & A_{12} \\ \dots & \dots & \dots \\ A_{21} & \vdots & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & \vdots & B_{12} \\ \dots & \dots & \dots \\ B_{21} & \vdots & B_{22} \end{bmatrix} \\ &= \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & \vdots & A_{11}B_{12} + A_{12}B_{22} \\ \dots & \dots & \dots \\ A_{21}B_{11} + A_{22}B_{21} & \vdots & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix} \end{aligned}$$

6.6 RANK OF A MATRIX

A number 'r' is said to be the rank of a matrix A, if

- there is at least one $(r \times r)$ sub-matrix of A whose determinant is not equal to zero; and
- the determinant of every $(r+1)$ -rowed square sub-matrix of A is zero.

In other words, rank of a matrix is equal to the order of the highest order non-singular square matrix contained in A.

For example, the rank of matrix $\begin{bmatrix} 5 & 2 \\ -2 & -3 \end{bmatrix}$ is 2; because this second order

matrix corresponds to the determinant $\begin{bmatrix} 5 & 2 \\ -2 & -3 \end{bmatrix}$, which is not zero, hence this given

2 x 2 matrix contain a non-singular matrix of order 2.

Rank of the matrix $\begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$ is 1; the highest order determinant

$\begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$ is zero, next to this would be of the 1st order, which is non-zero.

Again, the rank of $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is zero.

Note:

- i) The rank of a matrix cannot exceed the number of its rows or columns – whichever is less.
- ii) The rank of a column or row matrix is at most 1, unless the matrix is a null matrix, where the rank is always zero.
- iii) Rank of the transpose of a matrix is the same as the rank of that matrix.

Example 13: Determine the rank of the matrix $\begin{bmatrix} 2 & -1 & 3 & 1 & 5 \\ 3 & 2 & 0 & 2 & 1 \end{bmatrix}$, Here,

picking any sub-matrix of order 2, it can easily be shown that value of the determinant is non-zero, hence rank = 2.

The process of computing rank of a matrix may be simplified by applying row/column operations which do not change the values of the corresponding determinant, as discussed earlier.

$$\begin{bmatrix} 2 & -1 & 3 & 1 & 5 \\ 3 & 2 & 0 & 2 & 1 \end{bmatrix} \sim \begin{bmatrix} 0 & 0 & 0 & 1 & 5 \\ -1 & 4 & -6 & 2 & 1 \end{bmatrix}$$

by,

$$C_1' \rightarrow C_1 - 2C_4$$

$$C_2' \rightarrow C_2 + C_4$$

$$C_2' \rightarrow C_2 - C_4$$

$$\sim \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ -1 & 4 & -6 & 2 & -9 \end{bmatrix}, \text{ by, } C_5' \rightarrow C_5 - 5C_4$$

$$C_1'' \rightarrow 6C_1' - C_3'$$

$$\sim \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & -9 \end{bmatrix}, \text{ by, } C_2'' \rightarrow 6C_2' - 4C_3'$$

$$C_3'' \rightarrow 9C_3' - 6C_5'$$

$$\sim \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \text{ by } R_2' \Rightarrow R_2 - 2R_1$$

$$\sim [0 \ I_2], by, R_2 "$$

So, rank of the matrix is 2

6.7 APPLICATION OF MATRICES

Matrix Method

Inverse of a matrix can be used to solve a system of linear simultaneous

equations. Consider a system of n equations in n unknowns $x_1, x_2, x_3, \dots, x_n$ given below.

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n$$

This system of equations can be written in matrix form as

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad \dots (1)$$

Or $AX = B$,

$$\text{where } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

If $n = 3, x_1 = x, x_2 = y, x_3 = z$, then (1) reduces to

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\text{Or } AX = B \quad \dots (2)$$

$$\text{Or } X = A^{-1}B \quad \dots (3)$$

$$\text{Or } X = \frac{1}{|A|} (\text{adj}A)B \quad \left[\because A^{-1} = \frac{1}{|A|} (\text{adj}A) \right]$$

$$\text{Or } |A|X = (\text{adj}A)B \quad \dots (4)$$

If $|A| = 0$, then A^{-1} does not exist. In this case, the given system of equations either has no solution or infinitely many solutions. In this case, we find $(\text{adj}A)B$. If $(\text{adj}A)B = O$, then

system has infinitely many solutions

$\left[\begin{array}{l} \because \text{in this case, (4)} \Rightarrow 0X = (\text{adj}A)B \\ \text{which holds for all matrices } X, \text{ i.e.} \\ \text{for all real values of } x, y, z. \end{array} \right]$

Consistent system:

A system of equations is said to be consistent if there is either unique solution or infinite number of solutions.

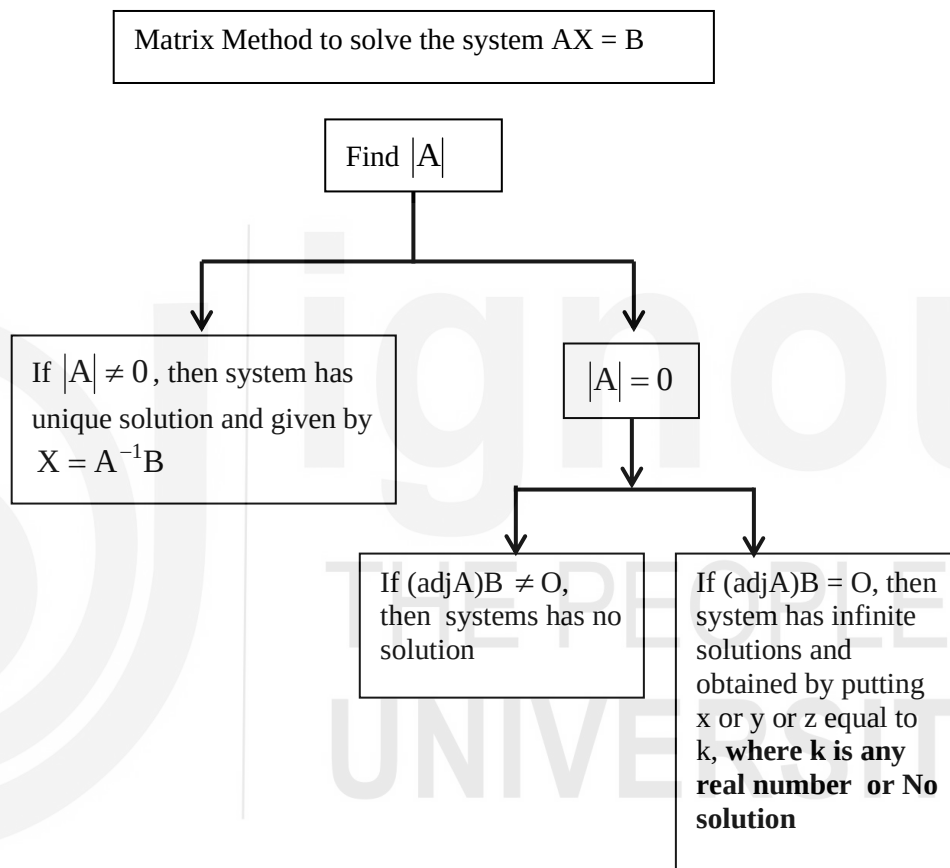
Inconsistent system:

If system has no solution then it is known as inconsistent system.

and if $(\text{adj}A)B \neq O$, then system has no solution.

$\left[\begin{array}{l} \because \text{in this case, L.H.S. of (4)} = O = \text{null matrix} \\ \text{but R.H.S. of (4)} = \text{non zero matrix.} \end{array} \right]$

Above discussion can be summarised in the following diagram.



Remark 8:

- (i) If $B = O$, i.e. if B is null matrix, then system given by (2) reduces to $AX = O$ and is known as linear homogeneous system of equations.
- (ii) Homogeneous system is always consistent.

Let us explain this method with the help of following example.

Example 14: The cost of 2 pens, 3 note-books, and 1 book is Rs 90. The cost of 1 pen, 4 note-books and 2 books is Rs 120. The cost of 2 pens, 4 note-books and 5 books is Rs 205. Find the cost of 1 pen, 1 note-book and 1 book by matrix method.

Solution:

Let Rs x, y, z be the cost of 1 pen, 1 note-book and 1 book respectively, then according to given

$$2x + 3y + z = 90$$

$$x + 4y + 2z = 120$$

$$2x + 4y + 5z = 205$$

In matrix form this system can be written as :

$$\begin{bmatrix} 2 & 3 & 1 \\ 1 & 4 & 2 \\ 2 & 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 90 \\ 120 \\ 205 \end{bmatrix}$$

$$\text{Or } AX = B \quad \dots (1)$$

$$|A| = \begin{vmatrix} 2 & 3 & 1 \\ 1 & 4 & 2 \\ 2 & 4 & 5 \end{vmatrix}$$

Expanding along R_1

$$|A| = 24 - 3 - 4 = 17 \neq 0$$

$\therefore A^{-1}$ exists.

Let A_{ij} denotes the cofactor of $(i, j)^{th}$ element of the matrix A.

$$\therefore A_{11} = (-1)^{1+1}(20-8) = 12$$

$$A_{12} = (-1)^{1+2}(5-4) = -1$$

$$A_{13} = (-1)^{1+3}(4-8) = -4$$

$$A_{21} = (-1)^{2+1}(15-4) = -11$$

$$A_{22} = (-1)^{2+2}(10-2) = 8$$

$$A_{23} = (-1)^{2+3}(8-6) = -2$$

$$A_{31} = (-1)^{3+1}(6-4) = 2$$

$$A_{32} = (-1)^{3+2}(4-1) = -3$$

$$A_{33} = (-1)^{3+3}(8-3) = 5$$

$$\text{adj}A = \begin{bmatrix} 12 & -1 & -4 \\ -11 & 8 & -2 \\ 2 & -3 & 5 \end{bmatrix}^T = \begin{bmatrix} 12 & -11 & 2 \\ -1 & 8 & -3 \\ -4 & -2 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}(\text{adj}A) = \frac{1}{17} \begin{bmatrix} 12 & -11 & 2 \\ -1 & 8 & -3 \\ -4 & -2 & 5 \end{bmatrix}$$

$$\text{Equation (1)} \Rightarrow X = A^{-1}B = \frac{1}{17} \begin{bmatrix} 12 & -11 & 2 \\ -1 & 8 & -3 \\ -4 & -2 & 5 \end{bmatrix} \begin{bmatrix} 90 \\ 120 \\ 205 \end{bmatrix}$$

$$= \frac{1}{17} \begin{bmatrix} 1080 - 1320 + 410 \\ -90 + 960 - 615 \\ -360 - 240 + 1025 \end{bmatrix} = \frac{1}{17} \begin{bmatrix} 170 \\ 255 \\ 425 \end{bmatrix} = \begin{bmatrix} 10 \\ 15 \\ 25 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 15 \\ 25 \end{bmatrix}$$

By definition of equality of two matrices, we have

$$x = 10, y = 15, z = 25$$

$\therefore$ costs of 1 pen, 1 note-book and one book are Rs 10, Rs 15, Rs 25, respectively.

Example 15: Solve the following system of equations:

$$4x + 2y = 6$$

$$6x + 3y = 8$$

Solution:

Given system of equations is

$$4x + 2y = 6 \quad \dots (1)$$

$$6x + 3y = 8 \quad \dots (2)$$

This system of equations can be written in matrix form as

$$\begin{bmatrix} 4 & 2 \\ 6 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

$$\text{Or } AX = B \quad \dots (3), \text{ where } A = \begin{bmatrix} 4 & 2 \\ 6 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, B = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 4 & 2 \\ 6 & 3 \end{vmatrix} = 12 - 12 = 0$$

$\Rightarrow$ system has either no solution or infinite many solutions.

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A.

$$\therefore A_{11} = (-1)^{1+1} (3) = 3, \quad A_{12} = (-1)^{1+2} (6) = -6$$

$$A_{21} = (-1)^{2+1} (2) = -2, \quad A_{22} = (-1)^{2+2} (4) = 4$$

$$\text{adj}A = \begin{bmatrix} 3 & -6 \\ -2 & 4 \end{bmatrix}^T = \begin{bmatrix} 3 & -2 \\ -6 & 4 \end{bmatrix}$$

$$(\text{adj}A)B = \begin{bmatrix} 3 & -2 \\ -6 & 4 \end{bmatrix} \begin{bmatrix} 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 18-16 \\ -36+32 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix} \neq O = \text{null matrix}$$

Hence system has no solution.

i.e. given system of equations is inconsistent.

Check Your Progress- 3

1. Solve the following system of equations:

$$3x + 6y - 4z = 3, \quad 3x - z = 0,$$

$$12x - 6y - z = -3$$

.....
.....
.....

2. Solve the following system of equations by matrix method:

$$(i) \ x + y = 3 \quad (ii) \ 2x - 3y = 3$$

$$4x - 3y = 5 \quad 6x - 9y = 5$$

.....
.....
.....

6.8 SUMMARY

In this unit explains concepts and applications of matrices, covering the fundamentals of Matrices, begins by defining a matrix and classifying its various types, followed by an in-depth exploration of core operations such as addition, subtraction, multiplication, and matrix inversion. This unit also introduces concepts like the adjoint, trace, and minors, alongside the rules governing the transpose operation of matrices. Subsequently, the unit identifies specific matrix forms, including orthogonal, symmetric, and idempotent matrices, before establishing the critical concept of the rank of a matrix and its relationship to Linear Dependence/Independence, finally the unit concluded with practical application of matrices.

6.9 SOLUTIONS/ANSWERS

Answer to Check Your Progress- 1

1.

(i) All the 8 elements can be arranged in single row, i.e. 1 row and 8 columns.

Or

(ii) They can be arranged in two rows with 4 elements in each row, i.e. 2 rows and 4 columns.

Or

(iii) In four rows with 2 elements in each row, i.e. 4 rows and 2 columns.

Or

(iv) In eight rows with 1 element in each row, i.e. 8 rows and 1 column.

∴ the possible orders are $1 \times 8, 2 \times 4, 4 \times 2, 8 \times 1$.

$$2. \quad A = [a_{ij}]_{3 \times 2} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}, \text{ where } a_{ij} = |i - j|$$

$$a_{11} = |1 - 1| = |0| = 0, a_{12} = |1 - 2| = |-1| = -(-1) = 1 \quad a_{21} = |2 - 1| = |1| = 1,$$

$$a_{22} = |2 - 2| = |0| = 0, a_{31} = |3 - 1| = |2| = 2, a_{32} = |3 - 2| = |1| = 1$$

$$\therefore A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 2 & 1 \end{bmatrix}$$

3.

Order	Type
-------	------

(i) 2×2	Square matrix [∵ rows and columns are equal in number.]
------------------	---

(ii) 2×2	Diagonal matrix [∵ all the non-diagonal elements are zero.]
-------------------	---

(iii) 2×2	Scalar matrix [∵ all the diagonal elements are equal and non diagonal element, are zero.]
--------------------	---

- (iv) 2×2 Identify matrix [$\because$ all the diagonal elements are unity and non diagonal element are zero.]
- (v) 3×3 Upper triangular matrix [$\because$ all the elements below the principal diagonal are zero.]
- (vi) 3×3 Lower triangular matrix [$\because$ all the elements above the principal diagonal are zero.]
- (vii) 3×1 Column matrix [$\because$ it has only one column.]
- (viii) 1×4 Row matrix [$\because$ it has only one row.]
- (ix) 2×3 Rectangular matrix
[$\because$ number of rows $\neq$ numbers of columns.]

4.

(i) We know that two matrices A and B are equal if :

- (a) their orders are same, and
- (b) the corresponding elements of A and B are equal.

$\therefore$ on comparing corresponding elements of two matrices, we have

$$3 = 3$$

$$x + y = 6 \quad \dots (1)$$

$$xy = 8 \quad \dots (2)$$

$$7 + z = 4 \Rightarrow z = -3$$

$$\text{From (1), } y = 6 - x \quad \dots (3)$$

Putting y from (3) in (2), we get $x(6 - x) = 8$

$$\Rightarrow 6x - x^2 - 8 = 0 \Rightarrow x^2 - 6x + 8 = 0 \Rightarrow x^2 - 4x - 2x + 8 = 0$$

$$\Rightarrow x(x - 4) - 2(x - 4) = 0 \Rightarrow (x - 4)(x - 2) = 0 \Rightarrow x = 4, 2$$

When $x = 4$, $y = 6 - 4 = 2$ and when $x = 2$, $y = 6 - 2 = 4$

$\therefore x = 4, y = 2, z = -3$ or $x = 2, y = 4, z = -3$.

(ii) We know that two matrices A and B are equal if

- (a) their orders are same, and
- (b) the corresponding elements of A and B are equal.

$\therefore$ on comparing corresponding elements of two matrices, we have

$$a + 5 = 2 \Rightarrow a = -3$$

$$-2a = a + 9 \Rightarrow -3a = 9 \Rightarrow a = -3$$

$$b + 6 = 11 \Rightarrow b = 5$$

$$2c = c + 4 \Rightarrow c = 4$$

$$3b + 2 = -b + 22 \Rightarrow 4b = 20 \Rightarrow b = 5$$

$$x = 3 - x \Rightarrow 2x = 3 \Rightarrow x = \frac{3}{2}$$

$$y + 1 = 2 \Rightarrow y = 1$$

$$z + 3 = 2z + 3 \Rightarrow -z = 0 \Rightarrow z = 0$$

$$x + 2 = 5 - x \Rightarrow 2x = 3 \Rightarrow x = \frac{3}{2}$$

$$\therefore a = -3, b = 5, c = 4, x = \frac{3}{2}, y = 1, z = 0.$$

5. We know that two matrices A and B are equal if

- (a) their orders are same, and
- (b) the corresponding elements of A and B are equal.

$\therefore$ on comparing corresponding elements of two matrices, we have

$$3x - 2y = -1 \quad \dots (1)$$

$$z + w = 7 \quad \dots (2)$$

$$3z - w = 5 \quad \dots (3)$$

$$x + y = 3 \quad \dots (4)$$

Equation (1) + 2 $\times$ equation (4) gives

$$\begin{array}{rcl} 3x - 2y & = & -1 \\ 2x + 2y & = & 6 \\ \hline 5x & = & 5 \end{array}$$

$$\Rightarrow x = 1$$

Putting $x = 1$ in (4), we get

$$1 + y = 3 \Rightarrow y = 2$$

(2) + (3) gives.

$$4z = 12 \Rightarrow z = 3$$

Putting $z = 3$ in (2), we get

$$3 + w = 7 \Rightarrow w = 4$$

$$\therefore x = 1, y = 2, z = 3, w = 4.$$

Answer to Check Your Progress- 2

$$1. \quad 3X + 2Y = \begin{bmatrix} 4 & 13 \\ 18 & 13 \end{bmatrix} \quad \dots (1)$$

$$2X - 3Y = \begin{bmatrix} 7 & 0 \\ -1 & -13 \end{bmatrix} \quad \dots (2)$$

Equation (1) $\times 3 + 2 \times$ equation (2) gives

$$9X + 6Y + 4X - 6Y = 3 \begin{bmatrix} 4 & 13 \\ 18 & 13 \end{bmatrix} + 2 \begin{bmatrix} 7 & 0 \\ -1 & -13 \end{bmatrix}$$

$$13X = \begin{bmatrix} 12 & 39 \\ 54 & 39 \end{bmatrix} + \begin{bmatrix} 14 & 0 \\ -2 & -26 \end{bmatrix} = \begin{bmatrix} 26 & 39 \\ 52 & 13 \end{bmatrix}$$

$$\Rightarrow X = \frac{1}{13} \begin{bmatrix} 26 & 39 \\ 52 & 13 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \text{ [By scalar multiplication property]}$$

Putting this value of X in (1), we get

$$3 \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} + 2Y = \begin{bmatrix} 4 & 13 \\ 18 & 13 \end{bmatrix}$$

$$\Rightarrow 2Y = \begin{bmatrix} 4 & 13 \\ 18 & 13 \end{bmatrix} - 3 \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 13 \\ 18 & 13 \end{bmatrix} + \begin{bmatrix} -6 & -9 \\ -12 & -3 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ 6 & 10 \end{bmatrix}$$

$$\Rightarrow Y = \frac{1}{2} \begin{bmatrix} -2 & 4 \\ 6 & 10 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 3 & 5 \end{bmatrix} \text{ [By scalar multiplication property]}$$

$$\therefore X = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \text{ and } Y = \begin{bmatrix} -1 & 2 \\ 3 & 5 \end{bmatrix}.$$

2. (i) Order of A is 1×2 and order of B is 3×1 .

$\therefore$ number of columns in A $\neq$ number of rows in B.

$\Rightarrow AB$ is not defined.

(ii) Number of columns in A = number of rows in B = 1.

$\Rightarrow AB$ is defined and is given by

$$AB = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \end{bmatrix} = \begin{bmatrix} 3 \times 5 & 3 \times 6 \\ 4 \times 5 & 4 \times 6 \end{bmatrix} = \begin{bmatrix} 15 & 18 \\ 20 & 24 \end{bmatrix}$$

(iii) AB is defined and is given by

$$AB = \begin{bmatrix} 3 & 4 & 1 \\ 2 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 3 + 12 + 5 & 6 + 16 + 6 \\ 2 + 15 + 30 & 4 + 20 + 36 \end{bmatrix} = \begin{bmatrix} 20 & 28 \\ 47 & 60 \end{bmatrix}$$

$$\begin{aligned} 3. [2 \quad 3 \quad 5] \begin{bmatrix} 4 & 5 \\ 0 & 1 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 \\ 5 & 6 & 8 \end{bmatrix} &= [8 + 0 + 10 \quad 10 + 3 + 30] \begin{bmatrix} 2 & 4 & 1 \\ 5 & 6 & 8 \end{bmatrix} \\ &= [18 \quad 43] \begin{bmatrix} 2 & 4 & 1 \\ 5 & 6 & 8 \end{bmatrix} \end{aligned}$$

$$= [36 + 215 \quad 72 + 258 \quad 18 + 344]$$

$$= [251 \quad 330 \quad 362]$$

4.

$$A^2 = AA = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1+0 & 0+0 \\ 2+6 & 0+9 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 8 & 9 \end{bmatrix}$$

$$A^4 = A^2 A^2 = \begin{bmatrix} 1 & 0 \\ 8 & 9 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 8 & 9 \end{bmatrix} = \begin{bmatrix} 1+0 & 0+0 \\ 8+72 & 0+81 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 80 & 81 \end{bmatrix}$$

$$\begin{aligned} A^8 &= A^4 A^4 = \begin{bmatrix} 1 & 0 \\ 80 & 81 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 80 & 81 \end{bmatrix} \\ &= \begin{bmatrix} 1+0 & 0+0 \\ 80+6480 & 0+6561 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 6560 & 6561 \end{bmatrix} \end{aligned}$$

5. $A = \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix}$

Let A_{ij} denotes the cofactor of $(i, j)^{\text{th}}$ element of the matrix A.

$$\therefore A_{11} = (-1)^{1+1}(5) = 5, \quad A_{12} = (-1)^{1+2}(4) = -4$$

$$A_{21} = (-1)^{2+1}(-3) = 3, \quad A_{22} = (-1)^{2+2}(2) = 2$$

$$\therefore \text{adj}A = \begin{bmatrix} 5 & -4 \\ 3 & 2 \end{bmatrix}^T = \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & -3 \\ 4 & 5 \end{vmatrix} = 10 - (-12) = 10 + 12 = 22$$

$$A(\text{adj}A) = \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} 10+12 & 6-6 \\ 20-20 & 12+10 \end{bmatrix} = \begin{bmatrix} 22 & 0 \\ 0 & 22 \end{bmatrix}$$

$$= 22 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 22I_2 = |A|I_2 \quad \dots (1)$$

$$(\text{adj}A)A = \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 10+12 & -15+15 \\ -8+8 & 12+10 \end{bmatrix} = \begin{bmatrix} 22 & 0 \\ 0 & 22 \end{bmatrix}$$

$$= 22 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 22I_2 = |A|I_2 \quad \dots (2)$$

From (1) and (2), we get

$$A(\text{adj}A) = (\text{adj}A)A = |A|I_2$$

6. (i) $\text{tr}(A) = \text{sum of diagonal elements} = 8 + 6 = 14$

(ii) We know that in an identity matrix, all the diagonal elements are unity.

$$\therefore \text{tr}(I_2) = 1 + 1 = 2 \quad [\because I_2 \text{ is identity matrix of order } 2 \times 2].$$

Similarly,

$$\text{tr}(I_3) = 1 + 1 + 1 = 3, \text{tr}(I_n) = \underbrace{1 + 1 + 1 + \dots + 1}_{n \text{ times}} = n.$$

7. $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}, B = \begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix}$

$$|A| = \begin{vmatrix} 2 & 3 \\ -1 & 4 \end{vmatrix} = 8 + 3 = 11 \neq 0$$

$\therefore A^{-1}$ exists.

$$|B| = \begin{vmatrix} 1 & 4 \\ 5 & -2 \end{vmatrix} = -2 - 20 = -22 \neq 0$$

$\therefore B^{-1}$ exists.

$$AB = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} 2 + 15 & 8 - 6 \\ -1 + 20 & -4 - 8 \end{bmatrix} = \begin{bmatrix} 17 & 2 \\ 19 & -12 \end{bmatrix}$$

$$|AB| = \begin{vmatrix} 17 & 2 \\ 19 & -12 \end{vmatrix} = -204 - 38 = -242 \neq 0$$

$\therefore (AB)^{-1}$ exists.

Let A_{ij}, B_{ij}, C_{ij} denotes the cofactors of $(i, j)^{th}$ element of matrices A, B, AB respectively.

$$\therefore A_{11} = (-1)^{1+1}(4) = 4, \quad A_{12} = (-1)^{1+2}(-1) = 1$$

$$A_{21} = (-1)^{2+1}(3) = -3, \quad A_{22} = (-1)^{2+2}(2) = 2$$

$$B_{11} = (-1)^{1+1}(-2) = -2, \quad B_{12} = (-1)^{1+2}(5) = -5$$

$$B_{21} = (-1)^{2+1}(4) = -4, \quad B_{22} = (-1)^{2+2}(1) = 1$$

$$C_{11} = (-1)^{1+1}(-12) = -12, \quad C_{12} = (-1)^{1+2}(19) = -19$$

$$C_{21} = (-1)^{2+1}(2) = -2, \quad C_{22} = (-1)^{2+2}(17) = 17$$

$$\therefore \text{adj}A = \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix}' = \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix}, \text{adj}B = \begin{bmatrix} -2 & -5 \\ -4 & 1 \end{bmatrix}' = \begin{bmatrix} -2 & -4 \\ -5 & 1 \end{bmatrix}$$

$$\text{adj}(AB) = \begin{bmatrix} -12 & -19 \\ -2 & 17 \end{bmatrix}' = \begin{bmatrix} -12 & -2 \\ -19 & 17 \end{bmatrix}, A^{-1} = \frac{1}{|A|}(\text{adj}A)$$

$$= \frac{1}{11} \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix}$$

$$B^{-1} = \frac{1}{|B|}(\text{adj}B) = \frac{1}{-22} \begin{bmatrix} -2 & -4 \\ -5 & 1 \end{bmatrix} = \frac{-1}{22} \begin{bmatrix} -2 & -5 \\ -4 & 1 \end{bmatrix}'$$

$$(AB)^{-1} = \frac{1}{|AB|}(\text{adj}AB) = \frac{-1}{242} \begin{bmatrix} -12 & -2 \\ -19 & 17 \end{bmatrix} \quad \dots (1)$$

$$B^{-1}A^{-1} = \frac{-1}{22} \begin{bmatrix} -2 & -4 \\ -5 & 1 \end{bmatrix} \frac{1}{11} \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix}$$

$$= \frac{-1}{242} \begin{bmatrix} -2 & -4 \\ -5 & 1 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix} = \frac{-1}{242} \begin{bmatrix} -8-4 & 6-8 \\ -20+1 & 15+2 \end{bmatrix}$$

$$= \frac{-1}{242} \begin{bmatrix} -12 & -2 \\ -19 & 17 \end{bmatrix} \quad \dots (2)$$

From (1) and (2), we have

$$(AB)^{-1} = B^{-1}A^{-1}$$

8. $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}, B = \begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix}$

$$|A| = \begin{vmatrix} 2 & 3 \\ -1 & 4 \end{vmatrix} = 8 + 3 = 11 \neq 0$$

$\therefore A^{-1}$ exists.

$$|B| = \begin{vmatrix} 1 & 4 \\ 5 & -2 \end{vmatrix} = -2 - 20 = -22 \neq 0$$

$\therefore B^{-1}$ exists.

$$AB = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} 2+15 & 8-6 \\ -1+20 & -4-8 \end{bmatrix} = \begin{bmatrix} 17 & 2 \\ 19 & -12 \end{bmatrix}$$

$$|AB| = \begin{vmatrix} 17 & 2 \\ 19 & -12 \end{vmatrix} = -204 - 38 = -242 \neq 0$$

$\therefore (AB)^{-1}$ exists.

Let A_{ij}, B_{ij}, C_{ij} denotes the cofactors of $(i, j)^{th}$ element of matrices A, B, AB respectively.

$$\therefore A_{11} = (-1)^{1+1}(4) = 4, \quad A_{12} = (-1)^{1+2}(-1) = 1$$

$$A_{21} = (-1)^{2+1}(3) = -3, \quad A_{22} = (-1)^{2+2}(2) = 2$$

$$B_{11} = (-1)^{1+1}(-2) = -2, \quad B_{12} = (-1)^{1+2}(5) = -5$$

$$B_{21} = (-1)^{2+1}(4) = -4, \quad B_{22} = (-1)^{2+2}(1) = 1$$

$$C_{11} = (-1)^{1+1}(-12) = -12, \quad C_{12} = (-1)^{1+2}(19) = -19$$

$$C_{21} = (-1)^{2+1}(2) = -2, \quad C_{22} = (-1)^{2+2}(17) = 17$$

$$\therefore \text{adj}A = \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix}, \text{adj}B = \begin{bmatrix} -2 & -5 \\ -4 & 1 \end{bmatrix}$$

$$\text{adj}(AB) = \begin{bmatrix} -12 & -19 \\ -2 & 17 \end{bmatrix}, A^{-1} = \frac{1}{|A|}(\text{adj}A) = \frac{1}{11} \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix}$$

$$B^{-1} = \frac{1}{|B|}(\text{adj}B) = \frac{1}{-22} \begin{bmatrix} -2 & -4 \\ -5 & 1 \end{bmatrix} = \frac{-1}{22} \begin{bmatrix} -2 & -5 \\ -4 & 1 \end{bmatrix}$$

$$(AB)^{-1} = \frac{1}{|AB|}(\text{adj}AB) = \frac{-1}{242} \begin{bmatrix} -12 & -2 \\ -19 & 17 \end{bmatrix} \dots (1)$$

$$B^{-1}A^{-1} = \frac{-1}{22} \begin{bmatrix} -2 & -4 \\ -5 & 1 \end{bmatrix} \frac{1}{11} \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix}$$

$$= \frac{-1}{242} \begin{bmatrix} -2 & -4 \\ -5 & 1 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix} = \frac{-1}{242} \begin{bmatrix} -8-4 & 6-8 \\ -20+1 & 15+2 \end{bmatrix}$$

$$= \frac{-1}{242} \begin{bmatrix} -12 & -2 \\ -19 & 17 \end{bmatrix} \dots (2)$$

From (1) and (2), we have

$$(AB)^{-1} = B^{-1}A^{-1}$$

Answer to Check Your Progress- 3

1.

Given system of equations is

$$3x + 6y - 4z = 3 \quad \dots (1)$$

$$3x - z = 0 \quad \dots (2)$$

$$12x - 6y - z = -3 \quad \dots (3)$$

This system of equations can be written in matrix forms as

$$\begin{bmatrix} 3 & 6 & -4 \\ 3 & 0 & -1 \\ 12 & -6 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix}$$

$$\text{Or } AX = B \quad \dots (4)$$

$$\text{where } A = \begin{bmatrix} 3 & 6 & -4 \\ 3 & 0 & -1 \\ 12 & -6 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 3 & 6 & -4 \\ 3 & 0 & -1 \\ 12 & -6 & -1 \end{vmatrix}$$

Expanding along R_1

$$|A| = 3(0-6) - 6(-3+12) - 4(-18-0) = -18 - 54 + 72 = 0$$

 $\therefore A^{-1}$ does not exist. $\Rightarrow$ system has either no solution or infinite many solutions.Let A_{ij} denotes the cofactor of $(i, j)^{th}$ element of the matrix A.

$$\therefore A_{11} = (-1)^{1+1}(0-6) = -6$$

$$A_{12} = (-1)^{1+2}(-3+12) = -9$$

$$A_{13} = (-1)^{1+3}(-18-0) = -18$$

$$A_{21} = (-1)^{2+1}(-6-24) = 30$$

$$A_{22} = (-1)^{2+2}(-3+48) = 45$$

$$A_{23} = (-1)^{2+3}(-18-72) = 90$$

$$A_{31} = (-1)^{3+1}(-6-0) = -6$$

$$A_{32} = (-1)^{3+2}(-3+12) = -9$$

$$A_{33} = (-1)^{3+3}(0-18) = -18$$

$$\text{adj}A = \begin{bmatrix} -6 & -9 & -18 \\ 30 & 45 & 90 \\ -6 & -9 & -18 \end{bmatrix} = \begin{bmatrix} -6 & 30 & -6 \\ -9 & 45 & -9 \\ -18 & 90 & -18 \end{bmatrix}$$

$$(\text{adj}A)B = \begin{bmatrix} -6 & 30 & -6 \\ -9 & 45 & -9 \\ -18 & 90 & -18 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix} = \begin{bmatrix} -18+0+18 \\ -27+0+27 \\ -54+0+54 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = O = \text{null matrix}$$

Hence system has infinite many solutions and given by putting either $x = k$ or $y = k$ or $z = k$ in any two equations given by (1), (2) and (3). Let us put $z = k$ in (1) and (2), we get, where k is any real number

$$3x + 6y = 3 + 4k \quad \dots (5)$$

$$3x = k \quad \dots (6)$$

$$\Rightarrow x = \frac{k}{3}$$

(6)

Putting $x = k/3$ in (5), we get

$$3(k/3) + 6y = 3 + 4k \Rightarrow 6y = 3 + 4k - k \Rightarrow 6y = 3 + 3k \Rightarrow y = (k+1)/2$$

$$\therefore x = \frac{k}{3}, y = \frac{k+1}{2}, z = k,$$

where k is any real number.

2.

(i) Given system of equations is

$$x + y = 3$$

$$4x - 3y = 5$$

This system of equations can be written in matrix form as

$$\begin{bmatrix} 1 & 1 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\text{Or } AX = B \quad \dots (1)$$

$$\text{where } A = \begin{bmatrix} 1 & 1 \\ 4 & -3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 1 \\ 4 & -3 \end{vmatrix} = -3 - 4 = -7 \neq 0$$

$\therefore A^{-1}$ exists.

Let A_{ij} denotes the cofactor of $(i, j)^{th}$ element of the matrix A .

$$\therefore A_{11} = (-1)^{1+1}(-3) = -3, \quad A_{12} = (-1)^{1+2}(4) = -4$$

$$A_{21} = (-1)^{2+1}(1) = -1, \quad A_{22} = (-1)^{2+2}(1) = 1$$

$$\text{adj}A = \begin{bmatrix} -3 & -4 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -3 & -1 \\ -4 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}(\text{adj}A) = -\frac{1}{7} \begin{bmatrix} -3 & -1 \\ -4 & 1 \end{bmatrix}$$

$$\Rightarrow X = A^{-1}B = -\frac{1}{7} \begin{bmatrix} -3 & -1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

Equation (1)

$$= -\frac{1}{7} \begin{bmatrix} -9-5 \\ -12+5 \end{bmatrix} = -\frac{1}{7} \begin{bmatrix} -14 \\ -7 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

By definition of equality of two matrices, we have

$$x = 2, y = 1$$

(ii) Given system of equations is

$$2x - 3y = 3$$

$$6x - 9y = 5$$

This system of equations in matrix form can be written as

$$\begin{bmatrix} 2 & -3 \\ 6 & -9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

Or $AX = B \dots (1)$,

where $A = \begin{bmatrix} 2 & -3 \\ 6 & -9 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$, $B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$

$$|A| = \begin{vmatrix} 2 & -3 \\ 6 & -9 \end{vmatrix} = -18 + 18 = 0$$

$\therefore A^{-1}$ does not exist.

$\Rightarrow$ system has either no solution or infinite many solutions.

Let A_{ij} denotes the cofactor of $(i, j)^{th}$ element of A .

$$\therefore A_{11} = (-1)^{1+1}(-9) = -9, A_{12} = (-1)^{1+2}(6) = -6$$

$$A_{21} = (-1)^{2+1}(-3) = 3, A_{22} = (-1)^{2+2}(2) = 2$$

$$adj A = \begin{bmatrix} -9 & -6 \\ 3 & 2 \end{bmatrix}' = \begin{bmatrix} -9 & 3 \\ -6 & 2 \end{bmatrix}$$

$$(adj A)B = \begin{bmatrix} -9 & 3 \\ -6 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} -27 + 15 \\ -18 + 10 \end{bmatrix} = \begin{bmatrix} -12 \\ -8 \end{bmatrix} \neq 0 = \text{nullmatrix}$$

Hence system has no solution.

6.9 REFERENCES AND FURTHER READINGS

1. BCS-012 : Basic Mathematics :
https://onlinecourses.swayam2.ac.in/nou26_ma07/preview (January 2026 session)
2. Mathematics Class XII Part – 1 By Ravi Prakash , GRB Books, Second Edition 2025